ITCS3166
Computer Networks

Lecture 3 Data Link Layer

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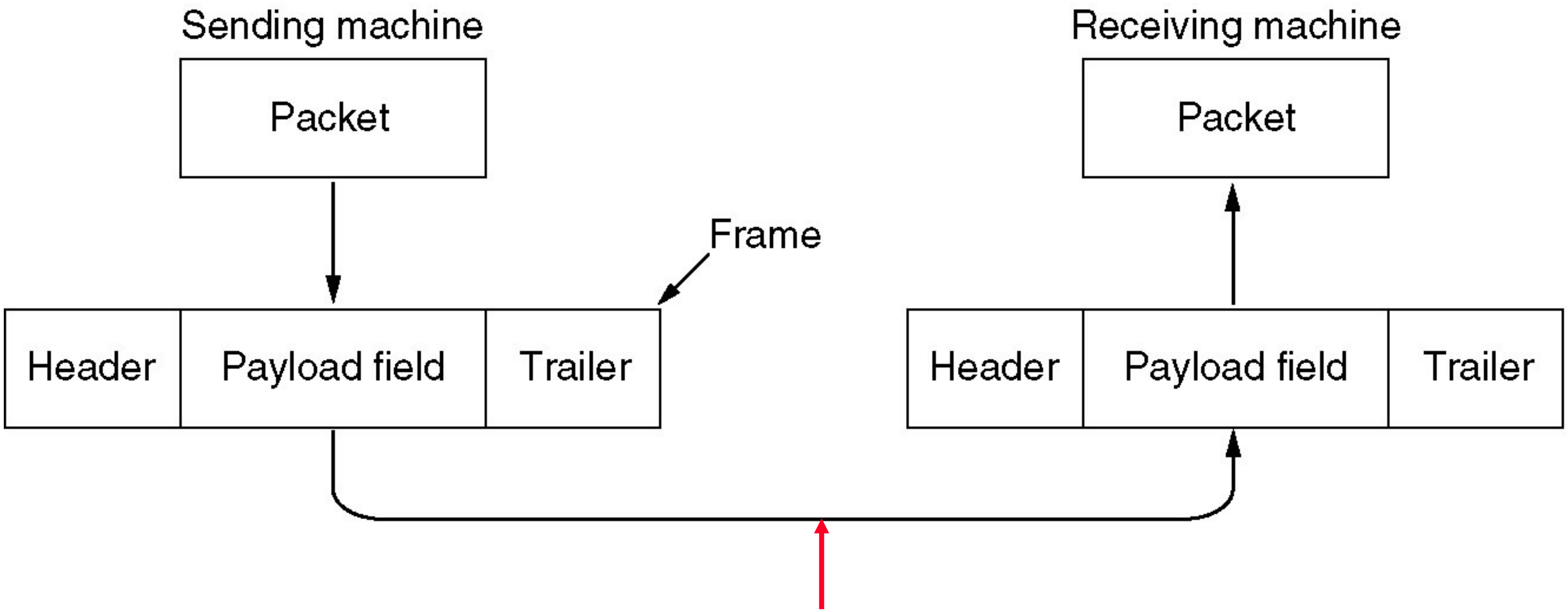
Data Link Layer Design Issues

- Services Provided to the Network Layer
  - Provide service interface to the network layer
- Framing
- Error Control
  - Dealing with transmission errors
- Flow Control
  - Slow receivers not swamped by fast senders

Can error control and flow control be done at other layers, e.g., the transport layer?

Functions of the Data Link Layer

Relationship between packets and frames.

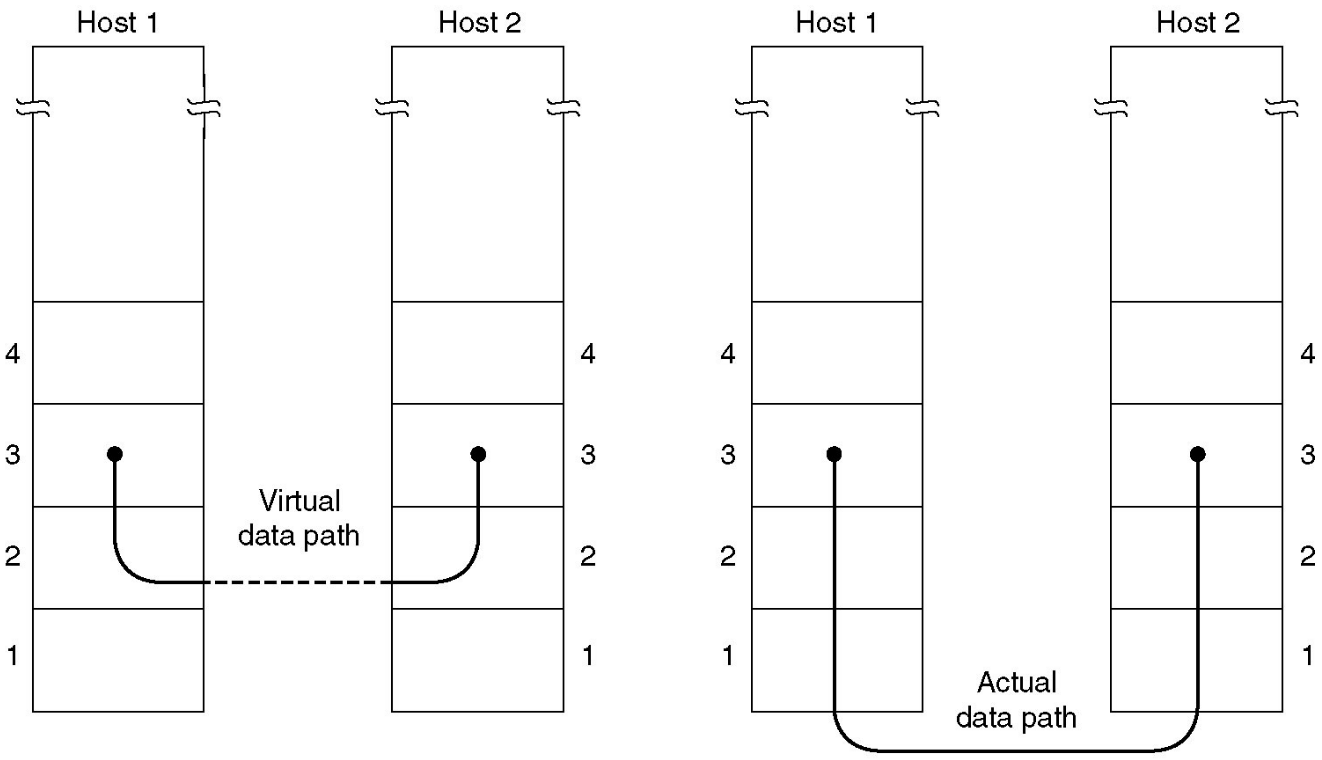


What is the essential property of a "wire-like" channel?

bits are delivered in order

Services Provided to Network Layer

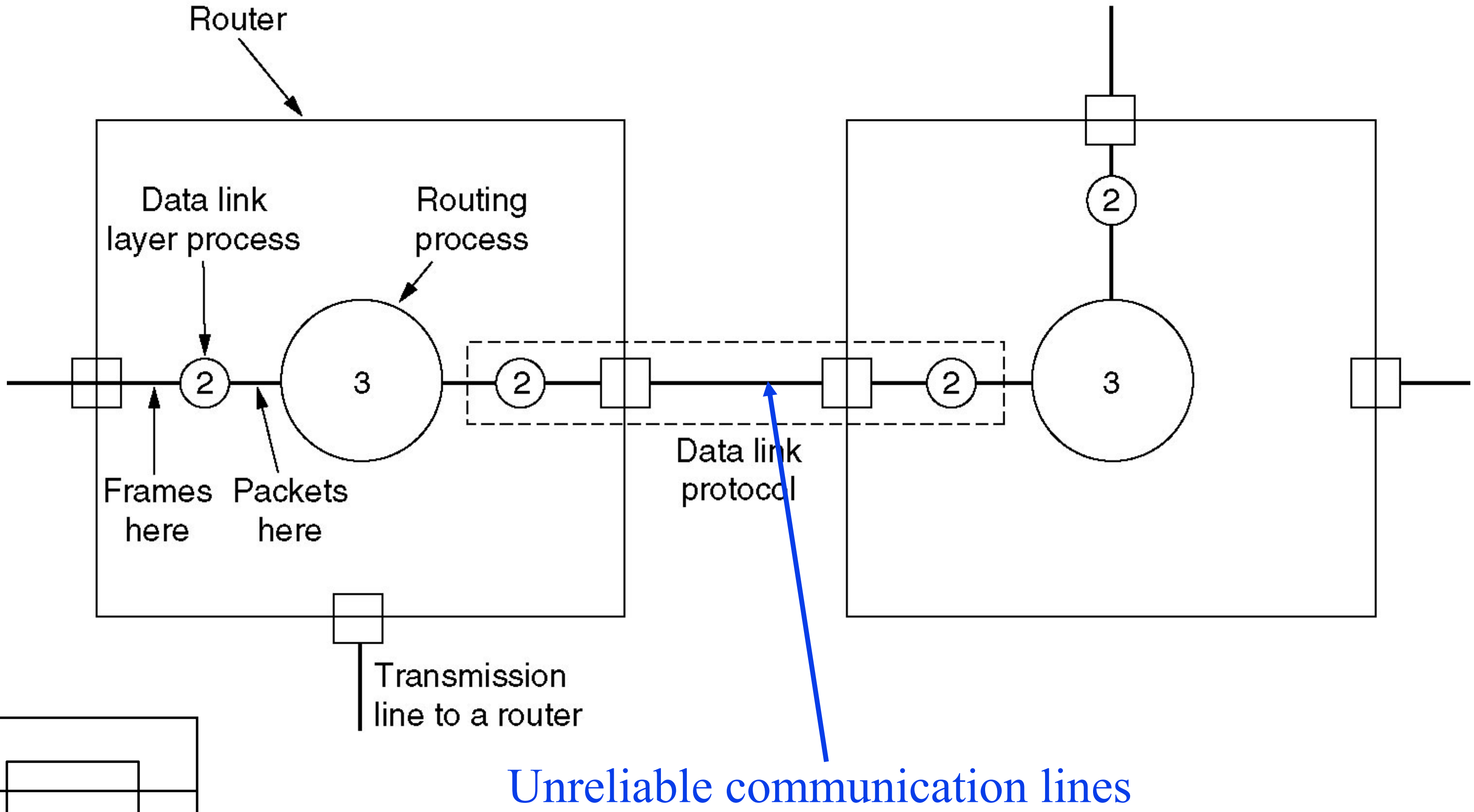
- Unacknowledged connectionless service
- Acknowledged connectionless service; say in wireless networks (optimization vs. requirement)
- Acknowledged connection-oriented service (no duplicate)



(a) Virtual communication. (b) Actual communication.

Services Provided to Network Layer (2)

Placement of the data link protocol.



Framing

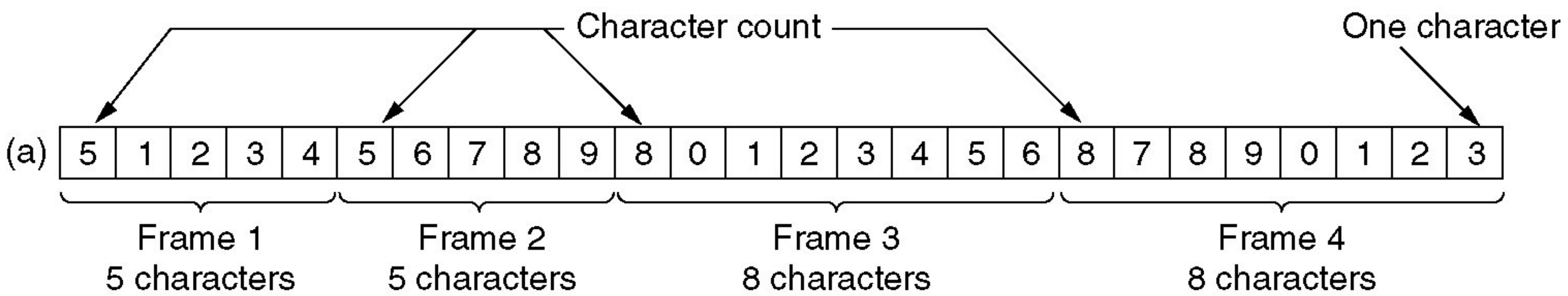
- Framing: to break the bit stream up into discrete frames and compute the checksum for each frame.

What is provided to the data link layer by the physical layer?

Why framing?

Framing – Character Count

- How to break the bit stream up into discrete frames?
- Character count: use a field in the header to specify the number of characters in the frame
- Example: code (01100011 01101111 01100100 01100101)



What is the hexadecimal value of "c"?/Binary/Octal?

What is the maximum size of a frame?

What is the synchronization trouble?

Answer

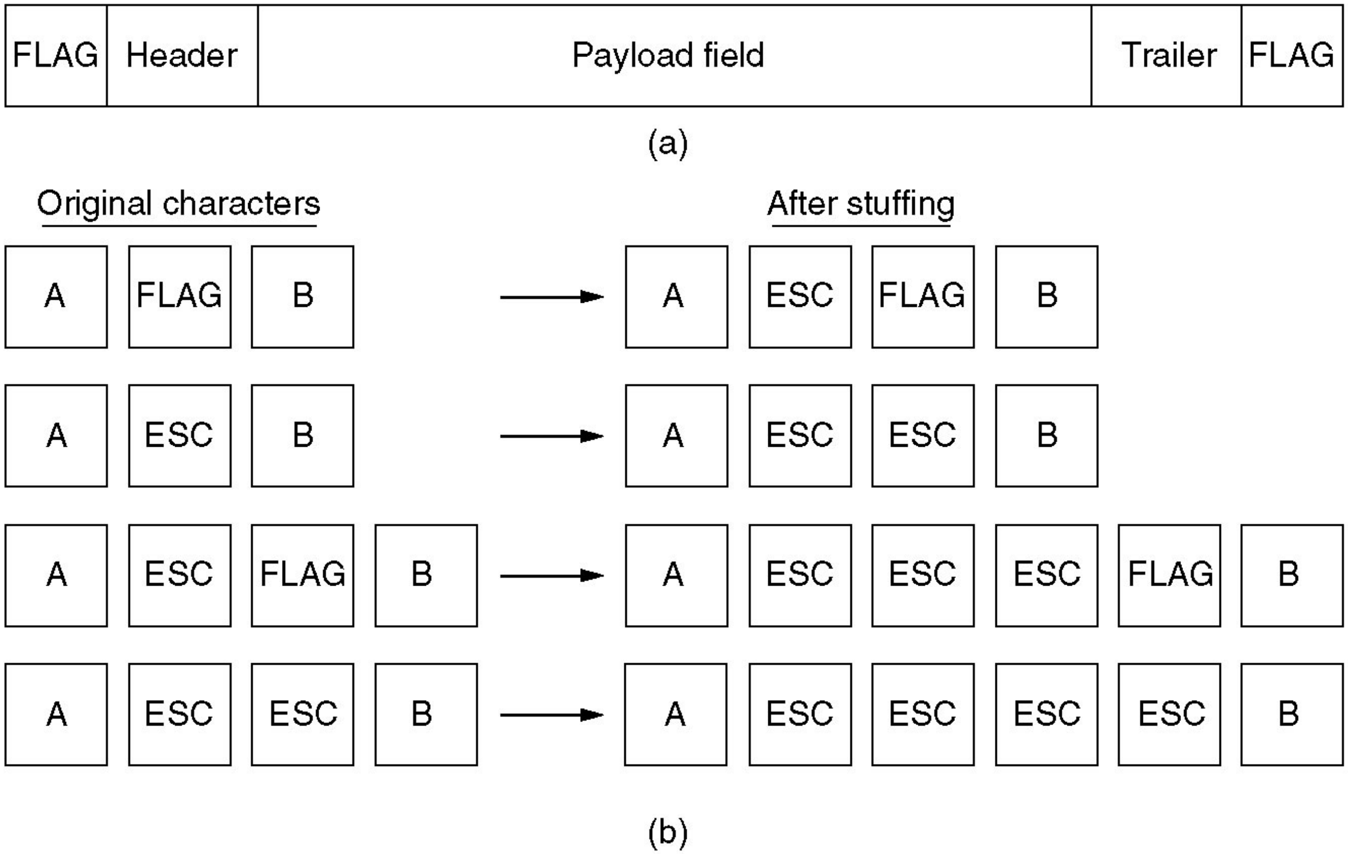
Hex	Binary	Octal
0	0	0
1	1	1
2	10	2
3	11	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000	10
9	1001	11
A	1010	12
B	1011	13
C	1100	14
D	1101	15
E	1110	16

The original Ethernet IEEE 802.3 standard defined the minimum Ethernet frame size as **64 bytes** and the maximum as **1518 bytes**. The maximum was later increased to**1522 bytes** to allow for VLAN tagging. The minimum size of an Ethernet frame that carries an ICMP packet is **74 bytes**.

Framing- Flag Byte with Byte Stuffing

- **Flag byte starts and ends each frame (used in PPP).**

What happens if the payload field contains the flag byte?



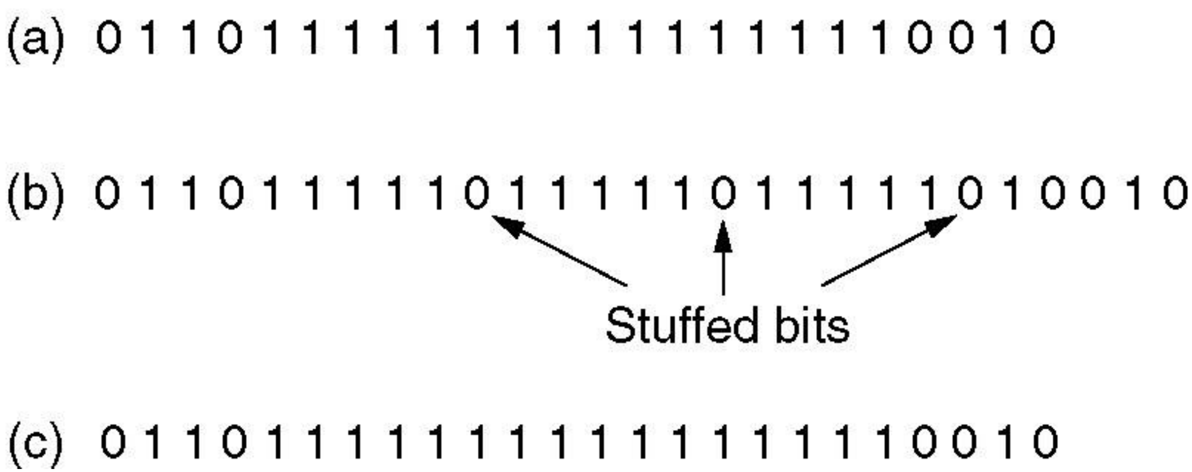
(a) A frame delimited by flag bytes.

(b) Four examples of byte sequences before and after stuffing.

Framing- Flag Byte with Bit Stuffing

- Problem with flag byte and byte stuffing: not all character codes use 8-bit characters.
- To allow a data frame with arbitrary number of bits and allow character codes with arbitrary number of bits per character
  - Flag byte: 01111110 (0X7E)
  - ESC: 0 (if 5 consecutive 1)

How to do de-stuffing?



Bit stuffing. (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in receiver's memory after de-stuffing.

Example 1

Q: **The following character encoding is used in a data link protocol:**

A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000

Show the bit sequence transmitted in binary for the four-character frame: A B ESC FLAG (space omitted)

(a) Character count

(b) Flag bytes with byte stuffing

(c) Starting and ending flag bytes, with bit stuffing

Answers

(a) Character count: 00000101 01000111 11100011 11100000 01111110

(b) Flag bytes with byte stuffing

01111110	01000111	11100011	11100000	11100000	11100000	01111110	01111110
flag	A	B	Esc (byte stuffed)	Esc	Esc (byte stuffed)	flag	flag

(c) Starting and ending flag bytes, with bit stuffing

01111110	01000111	110100011	111000000	011111010	01111110
flag	A	B	Esc	flag	flag

Question?

- One of your classmates has pointed out that it is wasteful to end each frame with a flag byte and then begin the next one with a second flag byte. One flag byte could do the job as well, and a byte saved is a byte earned. Do you agree?

Answer

- No. In a perfect situation when the data stream is continuous and there's no error, having only the starting flag byte is OK. However, if there is a long interval between the first frame and the 2nd frame, the receiver wouldn't be able to determine if the first frame has finished transmitting or not. Having both flags is also more stable.

Error Control: Error Detection and Correction

- What are errors?
  - Single errors vs. errors in burst
  - Advantages vs. disadvantages (if same BER)
    - Many 1-bit-error blocks vs. one 100-bit-error block

- Error-Correcting Codes

- Error-Detecting Codes

- What if a whole frame vanished or a whole packet is lost?
  - Flow control: Acknowledgement, retransmission, sequencing

Error Detection

Linear, systematic block codes

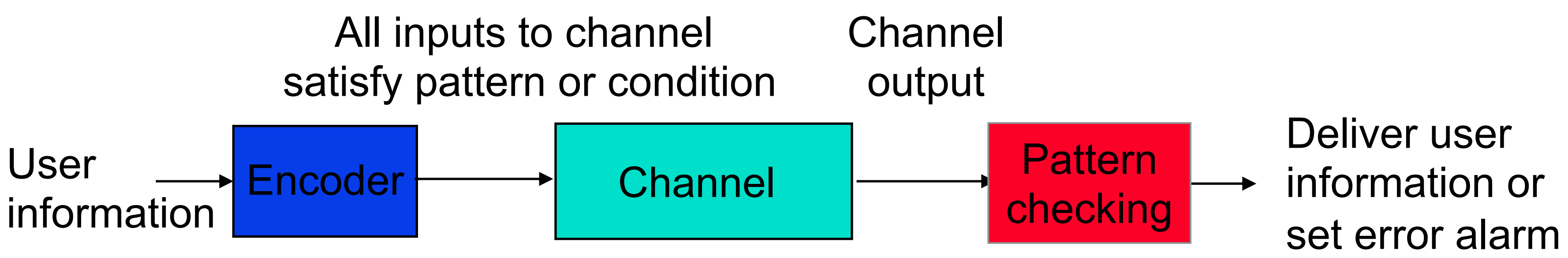
1. **Parity.**
2. **Checksums.**
3. **Cyclic Redundancy Checks (CRCs).**

Error Correction

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.
4. Low-Density Parity Check codes.

Key Idea

- All transmitted data blocks ("codewords") satisfy a pattern
  - If received block doesn't satisfy pattern, it is in error
  - Redundancy(r)
- Blindspot: when channel transforms a codeword into another codeword



Codeword and Hamming Distance

- A n-bit codeword: a frame of m-bit data plus r-bit redundant check bits ($n = m + r$)
- The number of bit positions in which two codewords differ is called the Hamming distance.
  - Example: 10001001 and 10110001 using XOR

Error Detecting Codes – Parity bit

- **Parity bit: to make the number of 1 bits in a codeword even or odd (r = 1)**
  - **Example: 10110100**

Can a parity bit used to detect a single-bit error in a codeword?

Can a parity bit used to detect a double-bit error in a codeword? Triple…?

Can a parity bit used to correct a single-bit error in a codeword?

Parity bit used in ASCII code

Error Correcting Codes – An Error-correcting Code

- **Given a complete list of the valid codewords, the minimum hamming distance of any two codewords is defined as the hamming distance of the complete code**

- **Example: a complete code with four legal codewords of 0000000000, 0000011111, 1111100000, 1111111111**

  What is the hamming distance of the code?

  How many error-bits at most can it correct?

  How many error-bits at most can it detect?

  What is the relationship between the hamming distance and the number of error-bits to be detected and corrected?

Error Correcting Codes – Low Limit on r

- A n-bit codeword: a frame of m-bit data plus r-bit redundant check bits ($n = m + r$)
- What is the lower limit on the number of bits needed to correct single-bit errors in a n-bit codeword?
  - $(n+1)\ 2^m <= 2^n$

Error Correcting Codes – Hamming Method

- A n-bit codeword: a frame of m-bit data plus r-bit redundant check bits (n = m + r)
- Use of a Hamming code to detect a& correct a single-bit error in a codeword
  - The bits that are powers of 2 are used as check bits.
  - The rest are filled up with the data bits
  - Each check bit forces the parity of some collection of bits, including itself, to be even (or odd)
  - To see which check bits the data bit in position k contributes to, rewrite k as a sum of powers of 2
  - A bit is checks by just those check bits occuring in its expansion (11 = 1 + 2 + 8)
- Example: a n-bit codeword containing a 7-bit data 1001000

  1001000 → 00110010000 (even-parity used)

  How to correct it if 00100010000 is received instead?

  How to correct it if 00110010001 is received instead?

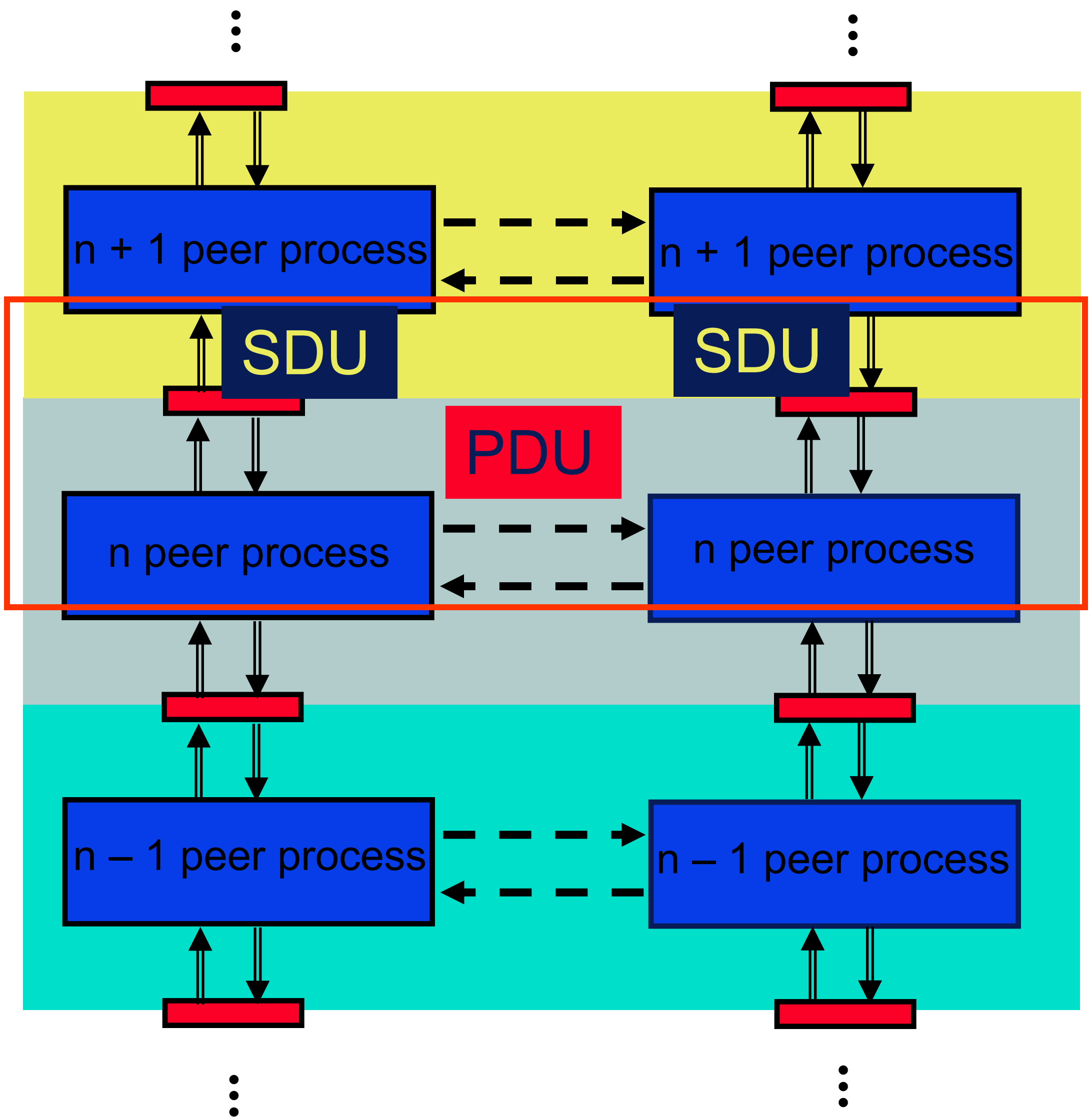
  How many check bits needed to d&c a single error in a 10-bit message

Error Detecting Codes vs. Error Correcting Codes

Why wireless networks prefer error correction while wired networks may go for error detection and retransmission?

What kind of applications prefer error correction instead of detection?

Data-Link Layer: Peer-to-Peer Protocols



- *Peer-to-Peer processes* execute layer-n protocol to provide service to layer-(n+1)
- Layer-(n+1) peer calls layer-n and passes Service Data Units (SDUs) for transfer
- Layer-n peers exchange Protocol Data Units (PDUs) to effect transfer
- Layer-n delivers SDUs to destination layer-(n+1) peer

Service Models

- The *service model* specifies the information transfer service layer-n provides to layer-(n+1)
- The most important distinction is if the service is:
  - Connection-oriented
  - Connectionless
- Service model possible features:
  - Arbitrary message size or structure
  - Sequencing and Reliability
  - Timing and Flow control
  - Multiplexing
  - Privacy, integrity, and authentication

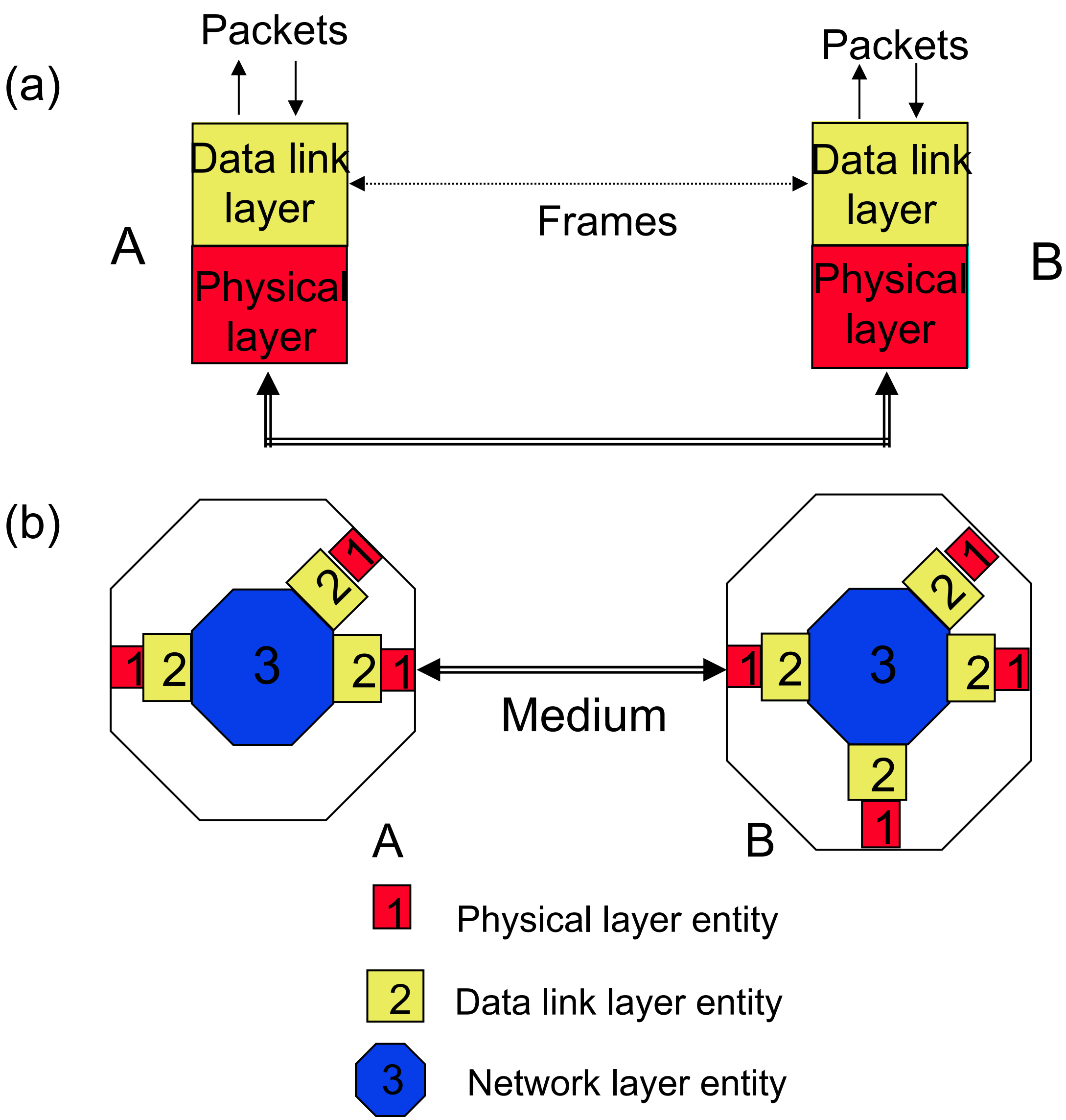
Reliability & Sequencing

- *Reliability*: what transmission is reliable?
  - *Sequencing*: Are messages or information stream delivered in order?
  - *duplication*
- How to provide reliable communication?
  - Examples: TCP and HDLC

Flow Control

- **What and Why flow control?**
  - **If destination layer-(n+1) does not retrieve its information fast enough, destination layer-n buffers may overflow**
- ***Flow Control* provide backpressure mechanisms that control transfer according to availability of buffers at the destination**
- **Examples: TCP and HDLC**

Error Control in Data Link Layer



- **Data Link operates over *wire-like*, directly-connected systems**
- **Frames can be corrupted or lost, but arrive in order**
- **Data link performs error-checking & retransmission**
- **Ensures error-free packet transfer between two systems**

Questions

- The following data fragment occurs in the middle of a data stream for which the byte-stuffing algorithm described in the text is used: A B ESC C ESC FLAG FLAG D. What is the output after stuffing?

After stuffing, we get A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D

Questions?

- What is the maximum overhead in byte-stuffing algorithm?

  - The maximum overhead occurs when the payload consists of only ESC and FLAG bytes. In this case, there will be 100% overhead.

Elementary Data Link Protocols

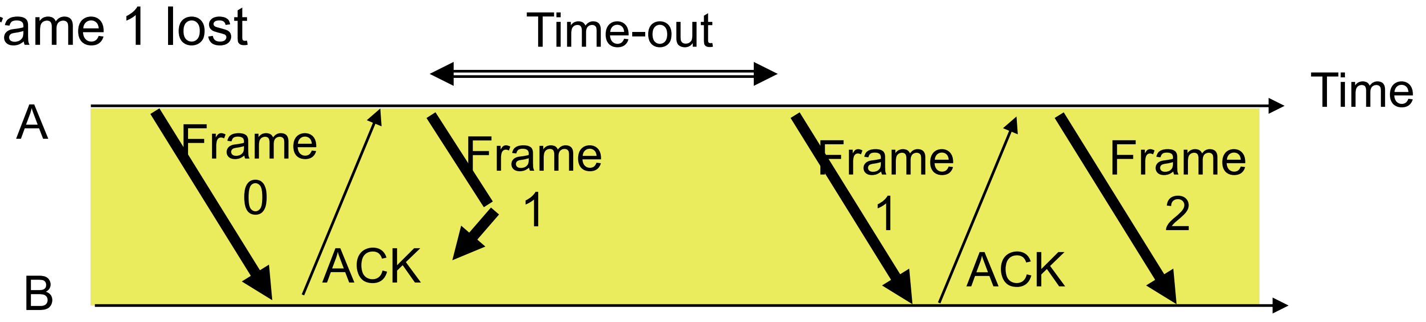
How to make sure all frames are eventually delivered to the network layer at the destination in the proper order?

- **An Unrestricted Simplex Protocol**
- **A Simplex Stop-and-Wait Protocol**
- **A Simplex Protocol for a Noisy Channel**

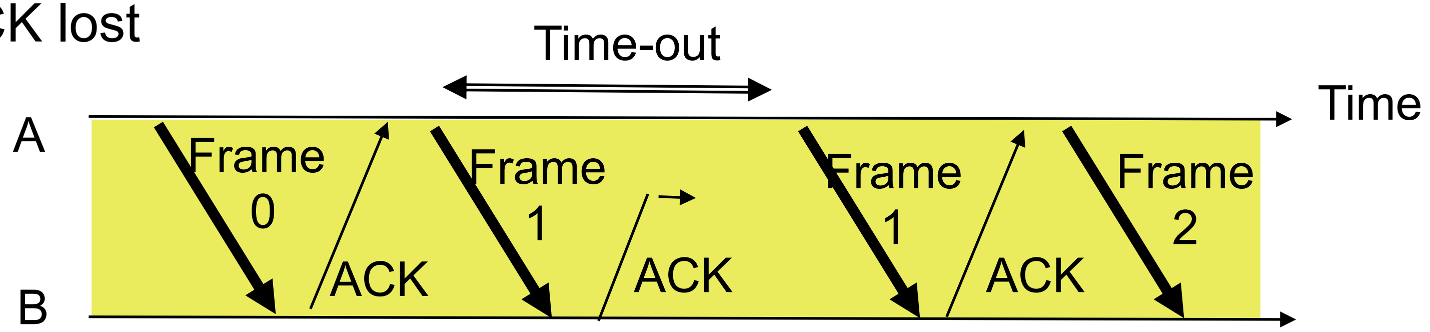
What if the Channel is Noisy?

Need for Sequence Numbers!

(a) Frame 1 lost



(b) ACK lost



- **In cases (a) & (b) the transmitting station A acts the same way**
- **But in case (b) the receiving station B accepts frame 1 twice**
- **Question: How is the receiver to know the second frame is also frame 1?**
- **Answer:** *Add frame sequence number in header*

Sliding Window Protocols (ARQ)

How about two-way data frame transmission?

- A One-Bit Sliding Window Protocol (Stop-and Wait)
- A Protocol Using Go Back N
- A Protocol Using Selective Repeat

***ARQ (Automatic Repeat rQquest) protocols* combine error detection, retransmission, and sequence numbering to provide reliability & sequencing**

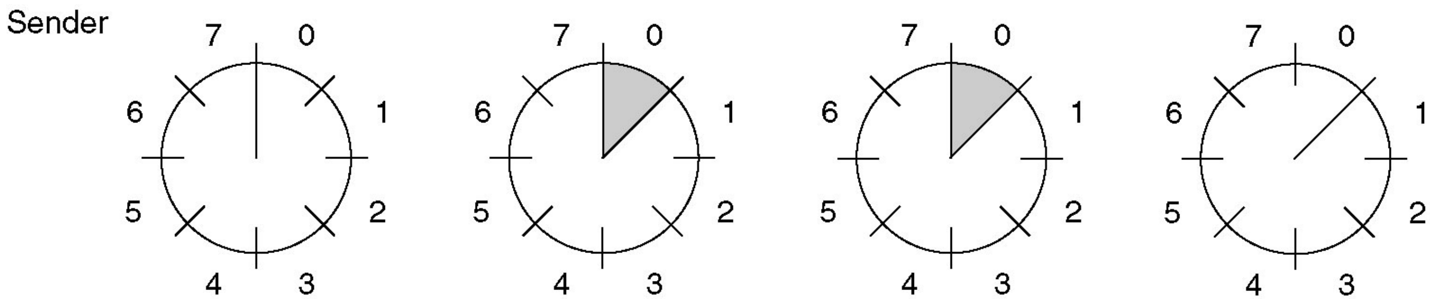
Sliding Window Protocols - Piggybacking

- Since in the two-way transmission, data frames and ACK frames are interleaving, why not have a free ride of ACK upon a data delivering?
  - How a receiver can tell if the frame is data or ACK?
- For piggybacking, how long should the data link layer wait for a packet onto which to piggyback the ACK?

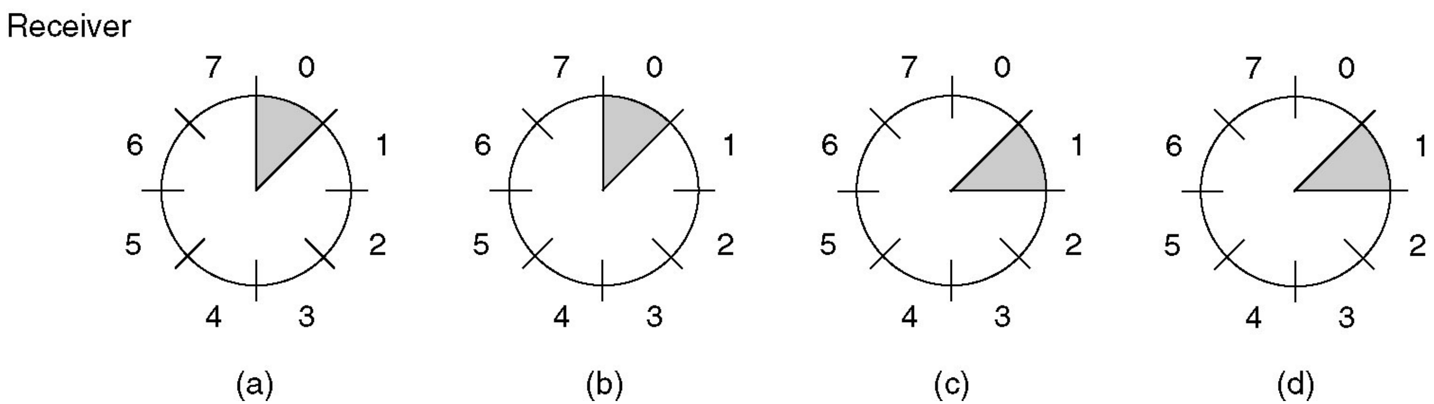
Metrics: Efficiency, Complexity, Buffer Requirements

Sliding Window Protocols – Sliding Window

- Sliding window: a set of Seq.# corresponding to frames permitted to send / receive.



What is the buffer size needed in the sender?



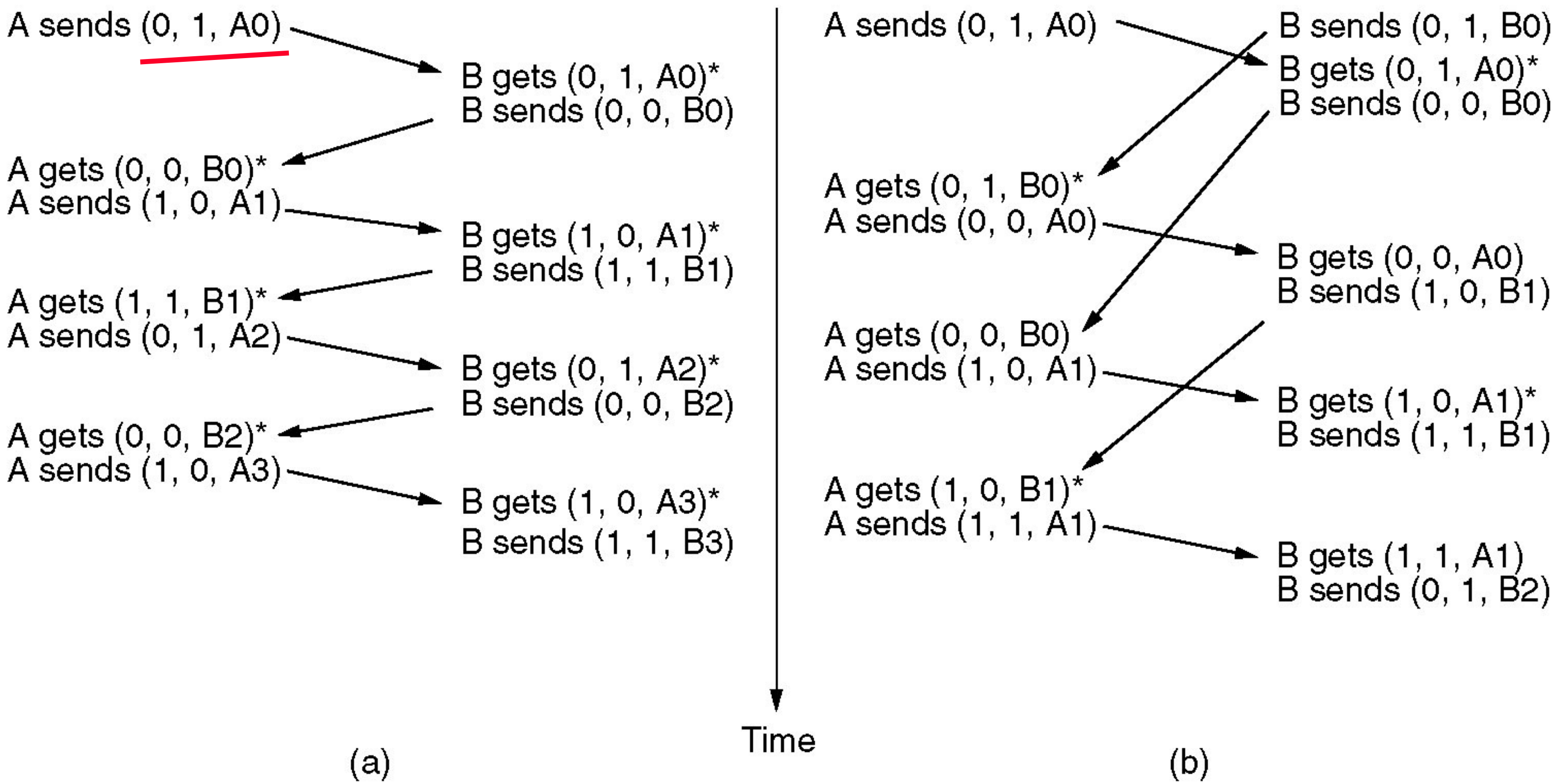
A sliding window of size 1, with a 3-bit sequence number.

(a) Initially. (b) After the first frame has been sent.

(c) After the first frame has been received.

(d) After the first acknowledgement has been received.

A One-Bit Sliding Window Protocol (2)



Two scenarios for protocol 4. (a) Normal case. (b) Abnormal case (simultaneous sending). The notation is (seq, ack, packet number). An asterisk indicates where a network layer accepts a packet.

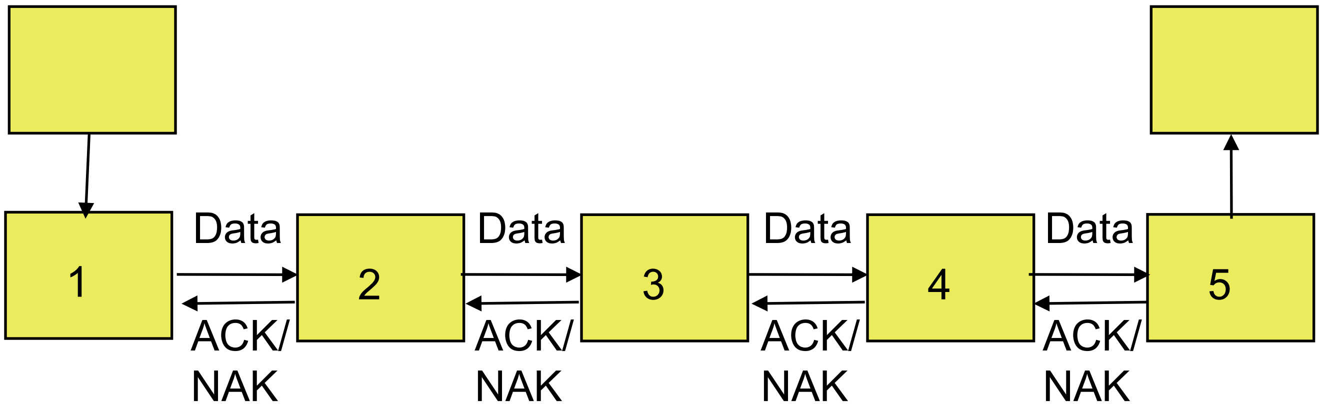
What is the problem with the case (b)?

End-to-End vs. Hop-by-Hop

- A service feature can be provided by implementing a protocol
  - end-to-end across the network
  - across every hop in the network
- Example:
  - Perform error control at every hop in the network or only between the source and destination?
  - Perform flow control between every hop in the network or only between source & destination?
- We next consider the tradeoffs between the two approaches

Which Approach Preferred

Hop-by-hop (HDLC)

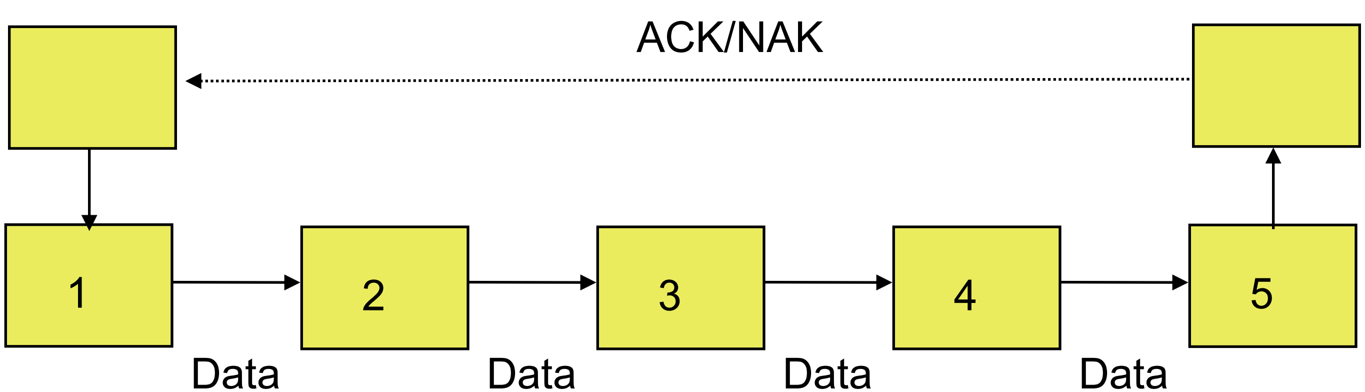


Hop-by-hop cannot ensure E2E correctness

Faster recovery

In situations with infrequent or frequent errors, which one?

End-to-end (TCP)



Simple inside the network

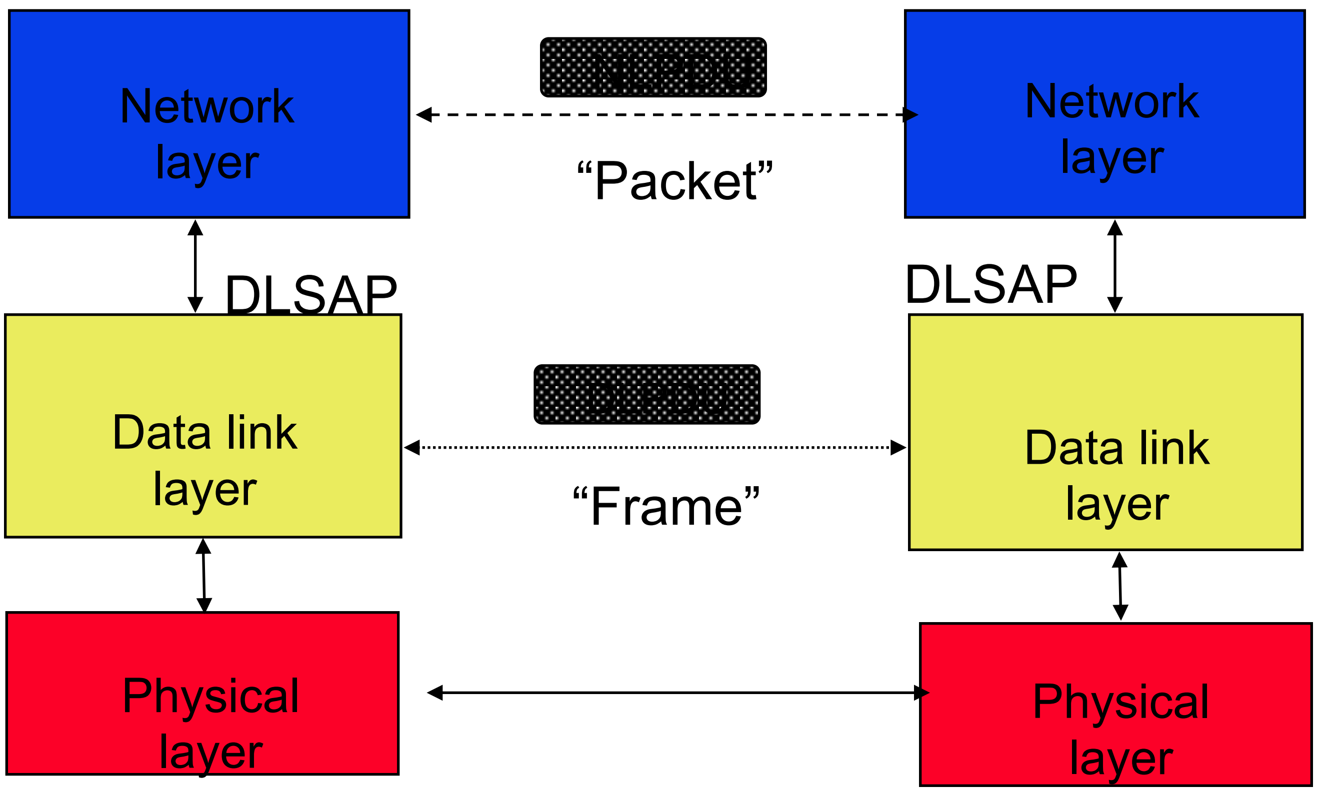
More scalable if complexity at the edge

Protocol Verification

- Finite State Machined Models
- Petri Net Models

Example 1: High-Level Data Link Control (HDLC)

- **Bit-oriented data link control**
  - **Derived from IBM Synchronous Data Link Control (SDLC)**



High-Level Data Link Control (HDLC)

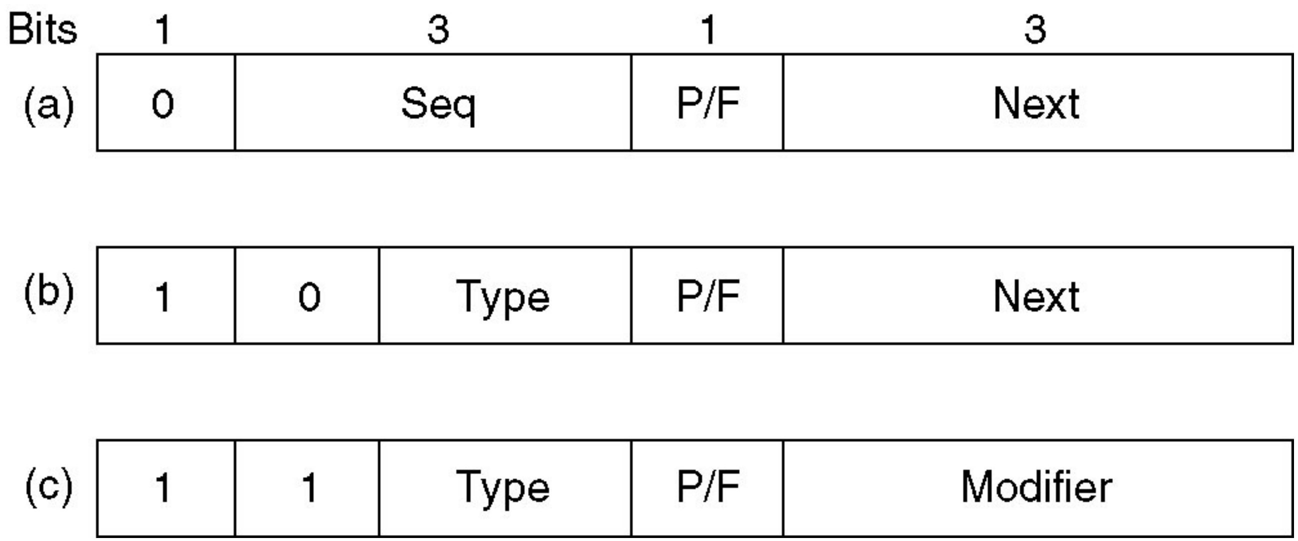
Frame format for bit-oriented protocols.

Bits	8	8	8	≥ 0	16	8
	0 1 1 1 1 1 1 0	Address	Control	Data	Checksum	0 1 1 1 1 1 1 0

Address: multi-drop or p2p (commands)
Control field gives HDLC its functionality, w/ 3bits for Sequencing
Checksum: $x^{16} + x^{12} + x^5 + 1$

What is the appropriate length for “Data” field?

High-Level Data Link Control (2)



Control field of

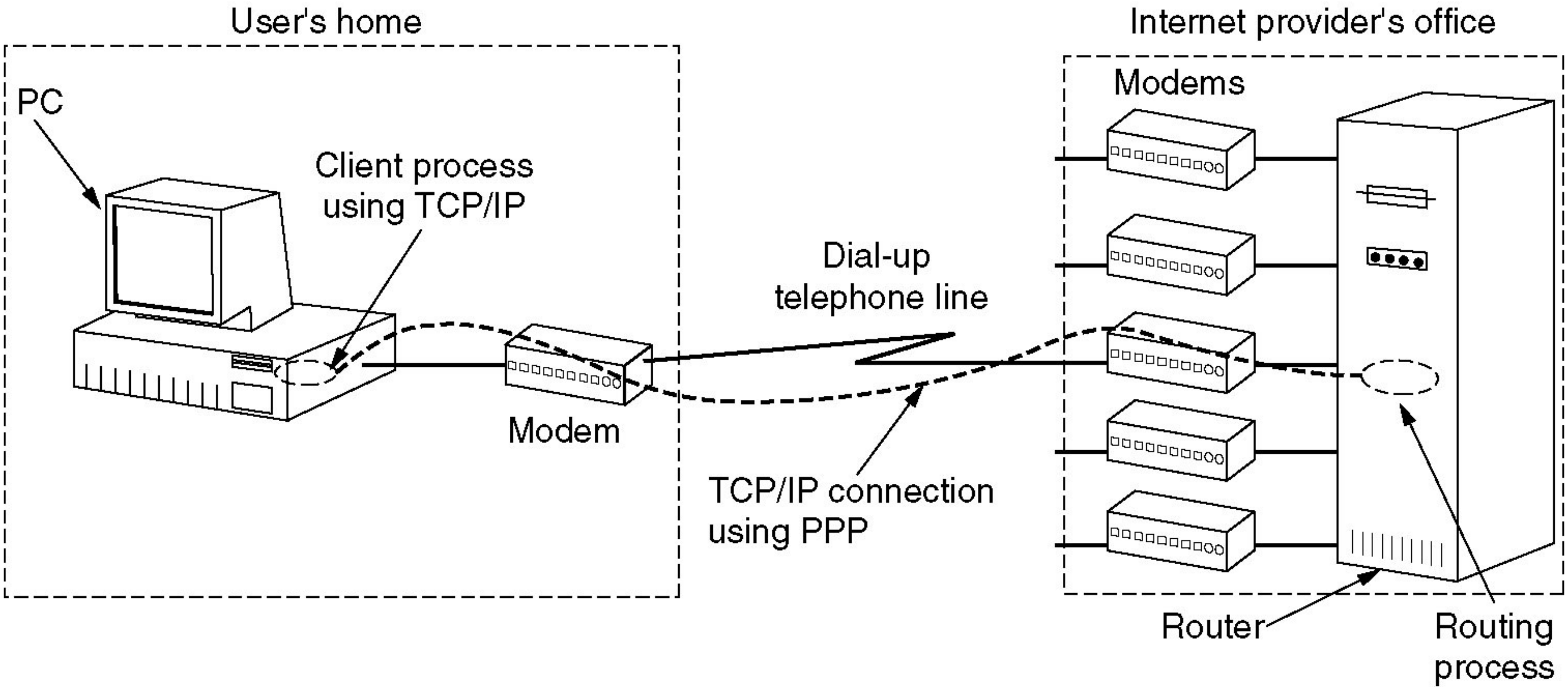
(a) An information frame.

(b) A supervisory frame.

(c) An unnumbered frame.

Example 2: The Data Link Layer in the Internet (PPP)

- Two situations that point-to-point communication is primarily used
  - Router-router leased line connection.
  - Dial-up host-router connections.



A home personal computer acting as an internet host.

PPP – Point to Point Protocol

- PPP provides three features
  - A method for framing, error detection, option negotiation, header compression, and optional reliable transmission
  - A link control protocol (LCP) for bringing lines up, testing, negotiating, and bring them down.
  - A way to negotiate network-layer options in a way that is independent of the network layer protocol to be used; e.g., NCP (Network Control Protocol).

Bytes	1	1	1	1 or 2	Variable	2 or 4	1
	Flag 01111110	Address 11111111	Control 00000011	Protocol	Payload	Checksum	Flag 01111110

The PPP full frame format for unnumbered mode operation.

What is the key difference between PPP framing and HDLC framing?

Bit-oriented vs. byte-oriented (character-oriented)

Reliability option vs. no reliability option

PPP Applications

PPP used in many point-to-point applications

- Telephone Modem Links
- Packet over SONET
  - IP→PPP→SONET
- PPP is also used over shared links such as Ethernet to provide LCP, NCP, and authentication features
  - PPP over Ethernet (RFC 2516)
  - Used over DSL

References

- Parity Check and Parity Bits (Error Detection): https://www.youtube.com/watch?v=jLuj62Gq-1I

- Error correction codes (Hamming coding): https://www.youtube.com/watch?v=cBBTWcHkVVY

- TCP: Sliding Windows: https://www.youtube.com/watch?v=klDhO9N01c4